Moment Generating Function

Consider a random variable X for which μ'_r exists, $r = 1, 2, 3, \ldots$. Then the generating function of the sequence $\{u'_r\}$ of moments should be defined as $\sum_{r=0}^{\infty} \frac{\mu'_r}{r!} t^r$ provided $\sum \frac{t^r}{r!}\mu'_r$ absolutely converges for $|t| < R$, for some $R > 0$. Then, $\sum \frac{t^r}{r!}\mu_r$ is a function of t say $M(t)$ and $M(t) = \sum \frac{t^r}{r!}\mu'_r, t < R$ is called the moment generating function of the RV X.

Note that $|t| < R$,

$$\begin{aligned} M_X(t) &= \sum_{r=0}^{\infty} \frac{t^r}{r!}\mu'_r \\ &= \sum_{r=0}^{\infty} \frac{t^r}{r!} E\left(X^r\right) \\ &= E\left[\sum_{r=0}^{\infty} \frac{t^r}{r!} x^r\right] = E\left[e^{tX}\right]. \end{aligned}$$

1 Definition

Let X be a RV such that $E\left(e^{tX}\right)$ exists, for $|t| < R$, for some $R > 0$, then the function of t,

$$M_X(t) = E\left(e^{tX}\right), |t| < R \quad \text{is called the MGF of } X.$$

Remark

1) Let $M(t)$ be the MGF of a RV X, then by definition,

$$\begin{aligned} M(t) = E\left[e^{tX}\right] &= E\left[\sum_{r=0}^{\infty} \frac{t^r}{r!} X^r\right] \\ &= \sum_{r=0}^{\infty} \frac{t^r}{r!} E\left(X^r\right) \\ &= \sum_{r=0}^{\infty} \frac{t^r}{r!}\mu'_r, \quad |t| < R. \end{aligned}$$

$\Rightarrow M(t)$ can be expressed as a convergent power series. By uniqueness, μ'_r can be obtained as the co-efficient of $\frac{t^r}{r!}$ in the expansion of $M(t)$ and this determination is unique. Given that $M_X(t)$ of a R.V X can generate the sequence of moments μ'_r that's why $M(t)$ is called moment generating function of X.

Differentiating the power series term by term,

$$\begin{aligned} \frac{\partial}{\partial t}\left[\sum_{r=0}^{\infty} \frac{t^r}{r!} u'_r\right] &= \sum_{r=0}^{\infty} \frac{\partial}{\partial t}\left\{\frac{t^r}{r!} u'_r\right\} \\ &= \sum_{r=1}^{\infty} \frac{t^{r-1}}{(r-1)!} u'_r \\ M^k(t) &= \sum_{r=k}^{\infty} \frac{t^{r-k}}{(r-k)!}\mu'_k \\ M'(0) &= M'(t)\big|_{t=0} = \mu'_1 \end{aligned}$$

$M''(0) = M''(t)|_{t=0} = \mu_2'$

In general,

$$M^{(p)}(0) = \mu_p' = E\left(X^p\right)$$

2) If MGF of a non-negative integer valued RV X exists and is $M(t)$,

$$M(t) = E\left[e^{tX}\right] = E\left[\left(e^t\right)^X\right] = P_X\left[e^t\right]$$

where $P_X(\cdot)$ is the PGF of X.

2 Exercise

Find the MGF of the following distribution.

i) $X \sim \text{Bin}(n, p)$

$$\begin{aligned} M_X(t) = E\left(e^{tx}\right) &= \sum_{x=0}^{n} e^{tx}\binom{n}{x}p^x q^{n-x} \\ &= \sum_{x=0}^{n}\binom{n}{x}\left(pe^t\right)^x q^{n-x} \\ &= \left(q + pe^t\right)^n \end{aligned}$$

ii) $X \sim \text{Poi}(\lambda)$

$$\begin{aligned} M_X(t) = E\left(e^{tX}\right) &= \sum_{x=0}^{\infty} e^{tx}\frac{e^{-\lambda}\lambda^x}{x!} \\ &= e^{-\lambda}\sum_{x=0}^{\infty}\frac{\left(\lambda e^t\right)^x}{x!} \\ &= e^{-\lambda}e^{\lambda e^t} = e^{-\lambda(1-e^t)} \end{aligned}$$

iii) $X \sim \text{NB}(r, q)$

$$\begin{aligned} M_X(t) = E\left(e^{tX}\right) &= \sum_{x=0}^{\infty} e^{tx}\binom{x+r-1}{x}p^r q^x \\ &= p^r\sum_{x=0}^{\infty}\binom{x+r-1}{x}\left(qe^t\right)^x \\ &= p^r\left(1-qe^t\right)^{-r}, \quad \text{if } qe^t < 1; \quad t < \ln\left(\frac{1}{q}\right) \end{aligned}$$

iv) If $X \sim \text{Rect}(a, b)$, find the MGF of X and hence find μ_r'.

$$\begin{aligned} M_X(t) &= \int_a^b e^{tx}\frac{1}{b-a}dx \\ &= \begin{cases} \frac{1}{b-a}\left[\frac{e^{tb}}{t} - \frac{e^{ta}}{t}\right], & t > 0 \\ 1 \quad , & t = 0 \end{cases} \\ &= \frac{1}{t(b-a)}\left[1 + tb + \frac{t^2b^2}{2!} + \frac{t^3b^3}{3!} + \cdots + \frac{t^rb^r}{r!} + \cdots\right. \\ &\quad \left. - \left(1 + ta + \frac{t^2a^2}{2!} + \frac{t^3a^3}{3!} + \cdots + \frac{t^ra^r}{r!} + \cdots\right)\right] \\ &= \frac{1}{(b-a)}\left[(b-a) + \frac{t}{2!}\left(b^2-a^2\right) + \frac{t^2}{3!}\left(b^3-a^3\right) + \cdots + \frac{t^{r-1}}{r^b}\left(b^r-a^r\right) + \ldots\right] \end{aligned}$$

Now, u_r' = coefficient of $\frac{t^r}{r!}$ in the expression of $M(t) = \frac{b^{r+1}-a^{r+1}}{(r+1)(b-a)}$.

Alternative, $M'(0) = \lim_{t\to 0} M'(t) = \lim_{t\to 0}\left[\frac{M(t)-M(0)}{t-0}\right]$

$$
\begin{aligned}
&= \lim_{t\to 0}\left[\frac{e^{tb}-e^{ta}}{t^2(b-a)} - 1\right] \\
&= \lim_{t\to 0}\left[\frac{e^{tb}-tb-\left(e^{ta}-ta\right)}{t^2(b-a)}\right] \\
&= \lim_{t\to 0}\frac{\frac{t^2\left(b^2-a^2\right)}{2!}+\frac{t^3\left(b^3-a^3\right)}{3!}+\cdots}{t^2(b-a)} \\
&= \left(\frac{b+a}{2}\right) = \mu_1'
\end{aligned}
$$

v) Find the MGF of Gamma (θ, n) distribution. Hence find its mean and varience.

$$
\begin{aligned}
M(t) &= \int_0^\infty e^{tx}\frac{e^{-x/\theta}x^{n-1}}{\theta^n \Gamma n}dx \\
&= \frac{1}{\theta^n \Gamma n}\int_0^\infty e^{-\frac{x}{\theta}(1-\theta t)}x^{n-1}dx \\
&= \frac{\Gamma n}{\frac{\left(\frac{1-\theta t}{\theta}\right)^n}{\theta^n\sqrt{n}}} = \frac{\theta^n}{\frac{}{(1-\theta t)^n}} \\
&= \frac{1}{(1-\theta t)^n} = (1-\theta t)^{-n}; (1-\theta t) > 0, t < \frac{1}{\theta}
\end{aligned}
$$

3 Properties

1. $X_1, X_2, \ldots, X_n \overset{iid}{\sim} f(x)$, and $S_n = \sum_{i=1}^n X_i$

   MGF of $S_n = M_{S_n}(t)$

$$
\begin{aligned}
&= E\left[e^{tS_n}\right] \\
&= E\left[e^{t[X_1+\ldots+X_n]}\right] \\
&= E\left[e^{tX_1}e^{tX_2}\ldots e^{tX_n}\right] \\
&= \prod_{i=1}^n E\left(e^{tx_i}\right), \quad X_1, X_2 \ldots X_n \text{ are independent, } e^{tX_1}, e^{tX_2}, \ldots, e^{tX_n} \text{ are also independent.} \\
&= \prod_{i=1}^n M_{X_i}(t) \\
&= \left[M_{X_1}(t)\right]^n, \quad \text{they are identically distributed.}
\end{aligned}
$$

2. If for a RV MGF exists, then it determines the distribution uniquely.

Problems

1. $X_i \sim \text{Exp}(\text{mean} = \theta)$, $\quad f_X(x) = \begin{cases} \frac{1}{\theta}e^{-x/\theta}, & x > 0, \theta > 0. \\ 0 & \text{otherwise.} \end{cases}$

$$
\begin{aligned}
M_X(t) &= \int_0^\infty e^{tx}\frac{e^{-x/\theta}}{\theta}dx \\
&= \frac{1}{\theta}\int_0^\infty e^{-x(1-t\theta)/\theta}dx = \frac{1}{\theta}\left[\frac{e^{-\frac{x}{\theta}(1-t\theta)}}{-\left(\frac{1-t\theta}{\theta}\right)}\right]_0^\infty \\
&= -\frac{1}{(1-t\theta)}\left[-e^0\right] = (1-t\theta)^{-1}
\end{aligned}
$$

Define, $S = \sum_{i=1}^{n} X_i$, then $M_S(t) = [M_{X_i}(t)]^n = (1-\theta t)^{-n}$ which is the MGF of a Gamma distribution. Hence,

$$\sum_{i=1}^{n} X_i \sim \text{Gamma}\,(\theta, n)$$

2. If the Gamma PDF looks like $\dfrac{\theta^n e^{-\theta x} x^{n-1}}{\Gamma n}$, $M_X(t) = \left(1 - \frac{t}{\theta}\right)^{-n}$.
$X_1 \sim \text{Exp(rate} = \theta) \equiv X_1 \sim \text{Exp}\,(\text{mean} = \frac{1}{\theta})$.

$$f_{X_i}(x) = \begin{cases} \theta e^{-\theta x_i}, & x > 0, \theta > 0 \\ 0, & \text{otherwise.} \end{cases}$$

$M_{X_1}(t) = \left(1 - \frac{t}{\theta}\right)^{-1} \Rightarrow S = \sum_{i=1}^{n} X_i$ will have the MGF $\left(1 - \frac{t}{\theta}\right)^{-n}$.
$\Rightarrow S \sim$ Gamma distribution with above PDF.

3. $X \sim \text{Gamma}(\theta, n_i), i = 1(1)\text{k}$, independently,

$$S = \sum_{i=1}^{k} X_i \sim \text{Gamma}\left(\theta, \sum_{i=1}^{k} n_i\right)$$

$$\begin{aligned} M_S(t) &= \prod_{i=1}^{k} (1-\theta t)^{-n_i} \\ &= (1-\theta t)^{-\sum_{i=1}^{k} n_i} \end{aligned}$$

4. If a RV is bounded, show that MGF of X exists.

$$\begin{aligned} & p[|X| < M] = 1 \text{ for some } M, \\ & tx \leqslant |t||x| \leqslant |t|M \text{ wp } 1 \\ \Rightarrow \quad & e^{tx} \leqslant e^{t|M|} \text{ wp } 1 \\ \Rightarrow \quad & E\left[e^{tX}\right] \leqslant e^{t|M|} \text{ wp } 1; \quad \text{if } M_X(t) \text{ exists} \end{aligned}$$

4 Central Moment Generating Function

If $M(t)$ be the MGF of a RV X with mean μ, the MGF of $(X - \mu)$ is,

$$\begin{aligned} M_{X-\mu}(t) &= E\left[e^{t(X-\mu)}\right] = e^{-t\mu} M_X(t) \\ &= E\left[\sum_{r=0}^{\infty} \frac{t^r}{r!}(X-\mu)^r\right] \\ &= \sum_{r=0}^{\infty} \frac{t^r}{r!}\mu_r \end{aligned}$$

Hence, μ_r is the coefficient of $\frac{t^r}{r!}$ in the expansion of $M_{X-r}(t) = e^{-t\mu} M_X(t)$.

Example

1. $X \sim N\left(\mu, \sigma^2\right)$. Find the MGF of X. Hence find $E(X-\mu)^r$.

$$\begin{aligned} M_X(t) &= \int_{-\infty}^{\infty} e^{tx} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx & \frac{x-u}{\sigma} = z \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{t(\sigma z + \mu)} e^{-\frac{1}{2}z^2} \sigma dz & \Rightarrow dx = \sigma dz \end{aligned}$$

$$
\begin{aligned}
&= \frac{e^{t\mu}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{t\sigma z - \frac{z^2}{2}} dz \\
&= \frac{e^{t\mu}}{\sqrt{2\pi}} e^{\frac{t^2\sigma^2}{2}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}\left(z^2 - 2t\sigma z + t^2\sigma^2\right)} dz \\
&= \frac{e^{t\mu}}{\sqrt{2\pi}} e^{\frac{t^2\sigma^2}{2}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z-t\sigma)^2} dz \\
&= e^{t\mu + \frac{t^2\sigma^2}{2}} \int_{-\infty}^{\infty} \underbrace{\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{z-t\sigma}{1}\right)^2}}_{N(t\sigma_n, 1)} dz \\
&= e^{t\mu + \frac{t^2\sigma^2}{2}}
\end{aligned}
$$

We know, $M_{x-\mu}(t) = e^{-t\mu}\mu_X(t) = e^{\frac{1}{2}t^2\sigma^2}$

$$
= \sum_{s=0}^{\infty} \frac{\left(\frac{1}{2}t^2\sigma^2\right)^s}{s!} = \sum_{s=0}^{\infty} \frac{t^{2s}(2s)!\sigma^{2s}}{2^s s!(2s)!}
$$

Note that, $M_{X-\mu}(t) = \sum_{r=0}^{\infty} \frac{t^r}{r!}\mu_r$
Hence, μ_n = The coefficient of $\frac{t^r}{r!}$ in the Expansion of $M_{X-\mu}(t)$

$$
= \begin{cases} \frac{(2s)!\sigma^{2s}}{2^s s!}, r = 2s \\ 0, \quad r = 2s-1 \end{cases} = \begin{cases} \sigma^{2s} 1.3.5 \ldots (2s-1) & , r = 2s \\ 0 & , r = 2s-1 \end{cases}
$$

2. $X_i \sim N\left(\mu_i, \sigma_i^2\right), i = 1(1)n$, independently. Find the MGF of $Y = \sum_{i=1}^{n} l_i X_i$ where l_i's are constants.

$$
\begin{aligned}
M_Y(t) = E\left[e^{tY}\right] &= E\left[e^{t[l_1X_1 + l_2X_2 + l_3X_3 + \ldots + l_nX_n]}\right] \\
&= E\left[e^{tl_1X_1} e^{tl_2x_2} \cdots e^{tl_nX_n}\right] \\
&= \prod_{i=1}^{n} E\left[e^{tl_iX_i}\right] = \prod_{i=1}^{n} M_{X_i}(l_it) \\
&= \prod_{i=1}^{n} e^{\left(tl_i\mu_i + \frac{t^2 l_i^2 \sigma_i^2}{2}\right)} \\
&= e^{t\sum_{i=1}^{n} l_i\mu_i + \frac{t^2}{2}\sum_{i=1}^{n} l_i^2\sigma_i^2}
\end{aligned}
$$

As we know if MGF exists, it determines the distribution of a RV uniquely.

$$
Y \backsim N\left(\sum_{i=1}^{n} l_i\mu_i, \sum_{i=1}^{n} l_i^2\sigma_i^2\right)
$$

If $l_i = \frac{1}{n}$. $\forall i$, $Y = \frac{1}{n}\sum_{i=1}^{n} X_i = \bar{X} \sim N\left(\mu, \sigma^2/n\right)$.

3. If $X \sim \Lambda\left(\mu, \sigma^2\right)$. Find $\mu_1' = E(X)$. Does MGF of X exist?

$$
Y = \log X \sim N\left(\mu, \sigma^2\right)
$$

$\mu_r' = E(X^r) = E\left(e^{Yr}\right) = M_Y(r) = e^{r\mu + \frac{r^2\sigma^2}{2}}$.
Here, for log normal distribution μ_r' exists for $r = 1, 2, 3, \ldots$.
So, we can construct the power series,

$$
\sum_{r=0}^{\infty} \frac{t^r}{r!}\mu_r = \sum_{r=0}^{\infty} \frac{t^r}{r!} e^{r\mu + \frac{r^2\sigma^2}{2}} = \sum_{r=0}^{\infty} a_r
$$

By ratio test, $l = \lim_{r\to\infty}\left|\frac{a_{r+1}}{a_r}\right| = \lim_{r\to\infty}\left\{\frac{|t|}{(r+1)}e^{\mu+\frac{(2r+1)\sigma^2}{2}}\right\}$

$$= e^{\mu+\frac{1}{2}\sigma^2}|t|\lim_{r\to\infty}\left(\frac{e^{r\sigma^2}}{r+1}\right) \to \infty$$

As $e^{r\sigma^2}$ increase more rapidy than $(r+1)$.

Hence, the power series diverges for $t \neq 0$. Therefore, the MGF of X does not exist.

The Lognormal distribution is an example of a distribution whose all moments exist but MGF does not.

4. Show that MGF of Cauchy distribution does not exist.

$$E\left(e^{tX}\right) = \int_{-\infty}^{\infty} e^{tx}\frac{1}{\pi\left(1+x^2\right)}dx$$

$$= \int_{-\infty}^{0} e^{tx}\frac{1}{\pi\left(1+x^2\right)}dx + \int_{0}^{\infty} e^{tx}\frac{1}{\pi\left(1+x^2\right)}dx = I_1 + I_2$$

for $t > 0, \quad e^{tx} > tx$,

$$I_2 = \int_0^{\infty} e^{tx}\frac{1}{\pi\left(1+x^2\right)}dx \geqslant \int_0^{\infty}\frac{tx}{\pi\left(1+x^2\right)}dx$$

$$\int_0^{\infty}\frac{tx}{\pi\left(1+x^2\right)}dx = \frac{t}{2\pi}\lim_{a\to\infty}\int_0^a \frac{2x}{1+x^2}dx.$$

$$= \frac{t}{2\pi}\lim_{a\to\infty}\left[\log\left|1+x^2\right|\right]_0^a \to \infty.$$

for, $t > 0, I_2 \to \infty$.

By Comparison test, the MGF of X does not exist.

Alternative,

If $x \sim \text{Cauchy}(0,1)$, $E|X| = \int_{-\infty}^{\infty}|x|\frac{1}{\pi(1+x^2)}dx > \infty$.

Hence, for $X \sim \text{Cauchy}(0,1)$, μ_1' does not exist and consequently $u_r', r \geqslant 2$, does not exist. Therefore we can not construct the power series $\sum_{r=0}^{\infty}\frac{t^r}{r!}\mu_r'$, which if converges gives us the MGF. Hence for $X \sim$ Cauchy$(0,1)$ the MGF does not exist.

5. Show that the MGF of the Pareto Distribution does not exist.

$$f(x) = \begin{cases} \frac{\theta {x_0}^{\theta}}{x^{\theta+1}} & , x > x_0 \\ 0, & ow \end{cases}$$

$$E\left(e^{tX}\right) = \int_{x_0}^{\infty} e^{tx}\frac{\theta x_0^{\theta}}{x^{\theta+1}}dx = \theta x_0^{\theta}\int_{x_0}^{\infty}\frac{e^{tx}}{x^{\theta+1}}dx$$

$\lim_{x\to\infty}\frac{e^{tx}}{x^{\theta}} = +\infty$ or e^{tx} increases more rapidly than x^{θ} as $x > \infty$. For $\varepsilon > 0$ there exists $a\delta > 0$ such that $\frac{e^{tx}}{x^{\theta}} > \epsilon$ whenever $x > \delta$.

$$\Rightarrow \frac{e^{tx}}{x^{\theta}x} > \frac{\epsilon}{x} \quad \forall x > \delta.$$

By comparison $\int_{x_0}^{\infty}\frac{e^{tx}}{x^{\theta+1}}dx > \epsilon\int_{x_0}^{\infty}\frac{1}{x}dx = \epsilon\lim_{R\to\infty}\int_{x_0}^{R}\frac{dx}{x} \to \infty$.

For $t > 0, E\left(e^{tX}\right) = \infty$, Hence the MGF of Pareto distribution does not exist.

6. Find the MGF of Laplace or Double Exponential Distribution.

$$X \sim \text{DE}(\mu, \sigma)$$

Then, $Y \sim \frac{X-\mu}{\sigma} \sim$ Laplace $(0,1)$, with the PDF,

$$f_Y(y) = \frac{1}{2}e^{-|y|}, \quad y \in \mathbb{R}$$

$$\begin{aligned}
M_Y(t) &= \int_{-\infty}^{\infty} e^{ty}\frac{1}{2}e^{-|y|}dy \\
&= \int_{-\infty}^{0} e^{ty}\frac{1}{2}e^{y}dy + \int_{0}^{\infty} e^{ty}\frac{1}{2}e^{-y}dy \\
&= \frac{1}{2}\left[\int_{-\infty}^{0} e^{y(t+1)}dy + \int_{0}^{\infty} e^{-y(1-t)}dy\right] \\
&= \frac{1}{2}\left[\frac{e^{y(t+1)}}{t+1}\bigg|_{-\infty}^{0} + \frac{e^{-y(1-t)}}{t-1}\bigg|_{0}^{\infty}\right] \\
&= \frac{1}{2}\left[\frac{1}{1+t} - \frac{1}{t-1}\right] \\
&= \frac{1}{2}\left[\frac{1}{1+t} + \frac{1}{1-t}\right]; \qquad 1+t>0, \quad 1-t>0 \\
&= \frac{1}{2}\frac{1-t+1+t}{(1+t)(1-t)} \\
&= \frac{1}{1-t^2}, |t|<1
\end{aligned}$$

So, the MGF of Y is $M_Y(t) = \frac{1}{(1-t^2)}, |t|<1$

$$\begin{aligned}
&X = \mu + \sigma y \\
&M_X(t) = E\left(e^{tX}\right) = E\left(e^{t(\mu+\sigma Y)}\right) = e^{t\mu}E\left(e^{t\sigma Y}\right) = e^{t\mu}M_Y\left(e^{t\sigma}\right) \\
&= \frac{e^{t\mu}}{1-t^2\sigma^2}; \quad |\sigma t|<1
\end{aligned}$$

7. Find the MGF of Logistic Distribution. Hence find its four central moments.

$$X \sim \text{Logistic}(0,1)$$

$$\begin{aligned}
f(x) &= \frac{e^x}{(1+e^x)^2} = \frac{e^{-x}}{(1+e^{-x})^2}, \quad x \in \mathbb{R} \\
M_X(t) &= \int_{-\infty}^{\infty} e^{tx}\frac{e^x}{(1+e^x)^2}dx \qquad y = \frac{1}{1+e^x} \Rightarrow y = \frac{e^{-x}}{1+e^{-x}} \\
&= \int_{1}^{0} -\left(\frac{1}{y}-1\right)^t dy \qquad \Rightarrow 1+e^x = \frac{1}{y} \Rightarrow e^x = \left(\frac{1}{y}-1\right) \\
&= \int_{0}^{1} \left(\frac{1}{y}-1\right)^t dy \qquad \Rightarrow dy = \frac{-e^x}{(1+e^x)^2}dx \\
&= \int_{0}^{1} y^{-t}(1-y)^t dy \\
&= \int_{0}^{1} y^{(1-t)-1}(1-y)^{t+1-1}dy = \text{Beta}_1((1-t),(1+t)) \\
&= \frac{\Gamma(1-t)\Gamma(1+t)}{\Gamma 2} = \Gamma(1-t)\Gamma(1+t) \\
&= t\Gamma(t)\Gamma(1-t), |t|<1
\end{aligned}$$

8. Show that the two distinct PDFs on $(0, \infty)$

$$f_1(x) = \frac{1}{x\sqrt{2\pi}} e^{-\frac{1}{2}(\ln x)^2}, \quad f_2(x) = f_1(x)(1 + \theta \sin(2\pi \log(x)); \quad -1 < \theta < +1$$

have the same moments. Does the sequence of moments uniquely determine the distribution?

Note that, $\int_0^\infty x^r f_2(x)dx = \int_0^\infty x^r f_1(x)\{1 + \theta \sin(2\pi \log x)\}dx$

$$= \int_0^\infty x^r f_1(x)dx + \int_0^\infty x^r f_1(x)\theta \sin(2\pi \log x)dx$$

Now,

$$\begin{aligned}
&\int_0^\infty x^r f_1(x)\theta \sin(2\pi \log x)dx \\
=&\int_0^\infty x^r \frac{\theta}{x\sqrt{2\pi}} e^{-\frac{1}{2}(\log x)^2} \sin(2\pi \log x)dx \\
=&\int_{-\infty}^\infty e^{ur} \frac{\theta}{\sqrt{2\pi}} e^{-\frac{1}{2}u^2} \sin(2\pi u)du \\
=&\theta \int_{-\infty}^\infty \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\left(u^2 - 2ru + r^2\right)} e^{\frac{r^2}{2}} \sin(2\pi u)du \\
=&\theta e^{r^2/2} \int_{-\infty}^\infty \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(u-r)^2} \sin(2\pi u)du \\
=&\theta e^{r^2/2} \underbrace{\int_{-\infty}^\infty \frac{e^{-\frac{1}{2}y^2}}{\sqrt{2\pi}} \sin(2\pi y)dy}_{\text{odd function}} \\
=&0
\end{aligned}$$

As,

$$u - r = y \Rightarrow du = dy$$
$$\sin(2\pi u) = \sin(2\pi r + 2\pi y) = \sin(2\pi y)$$

Hence,

$$\int_0^\infty x^r f_2(x)dx = \int_0^\infty x^2 f_1(x)dx$$

Both the PDF's have the same sequence of moments. Here, looking at the sequence of moments, we can't determine a distribution uniquely.

Justification

Consider a sequence $\{u'_r\}$ of moments. We can construct the power series $\Sigma \frac{t^r}{r_!} u'_r$, if $\Sigma \frac{t^r}{r_!} u'_r$ converges absolutely for $|t| < R, R > 0$, then $\sum \frac{t^r}{r!} u'_r$ is MGF of distribution with μ'_r as a sequence of moments. We know that MGF determines a probability distribution uniquely and hence if MGF exists of a distribution, then $\{u'_r\}$ the sequence of moments is unique.

But if MGF does not exist, then the sequence of moments is not unique.